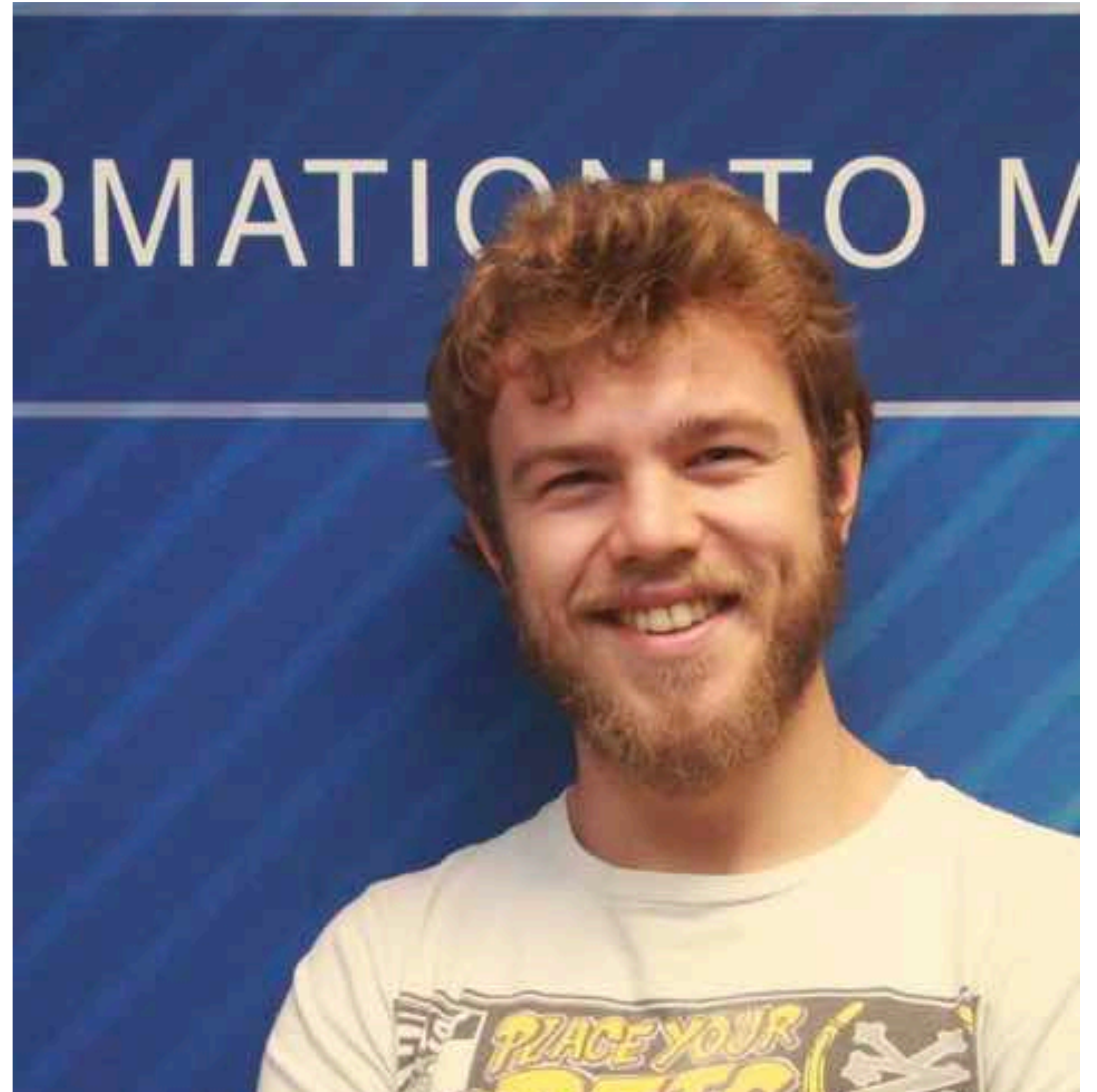


Thesis Proposals 2026

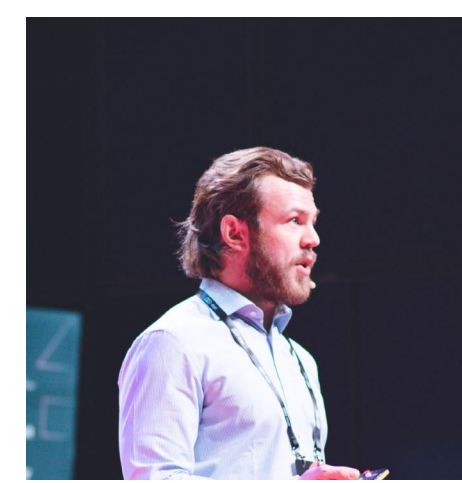
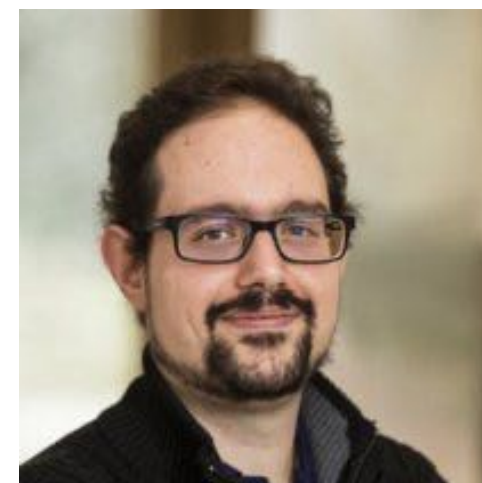
Who I Am



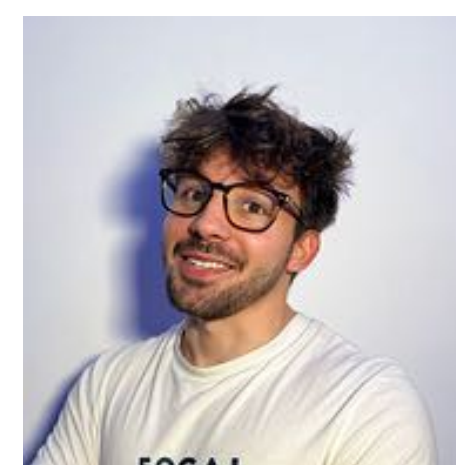
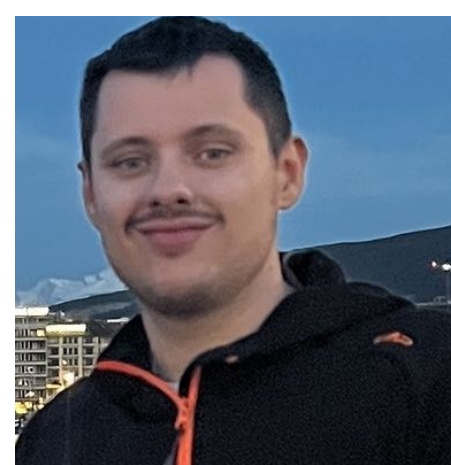
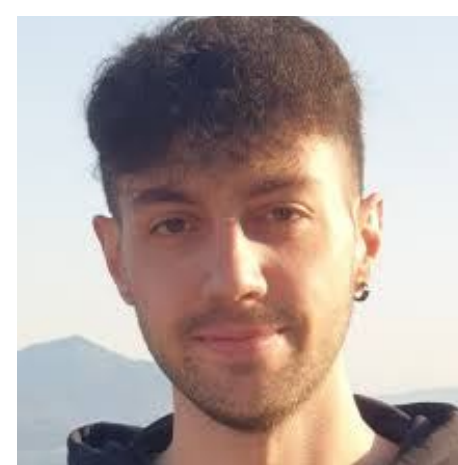
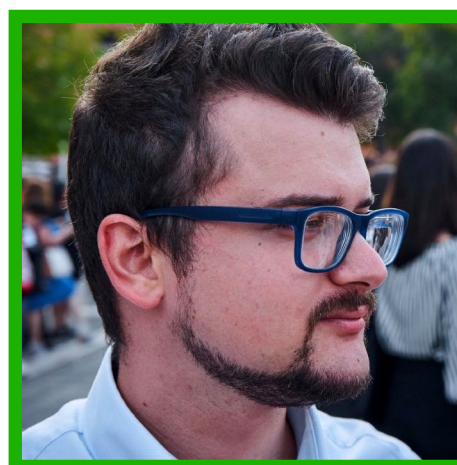
- **Riccardo Tommasini**
- Associate Professor
at INSA Lyon, LIRIS
- Former Lecturer at UT
- Streaming Enthusiast!
- 🇮🇹->🇺🇸->🇸🇪->🇫🇷



DB Team @



Prof. Angela Bonifati
DB Team's Head



Technical Details

Mauro Famá => Happy Intern now PhD Student

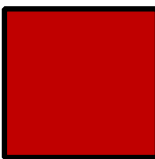
- 620 euro per months, max six month
- we student housing (FIFO)
 - about 200 french state contribution to the rent
 - train/bus ticket
- technical interview (with me)
- security clearance (takes about 2 months)
- riccardo.tommasini@insa-lyon.fr



Stream Processing

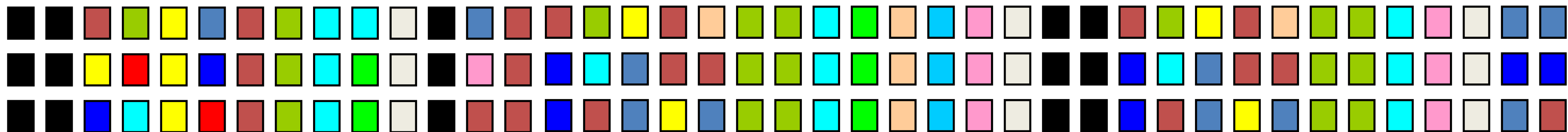
Intuition

How many red colored boxes are there?



??

infinite data



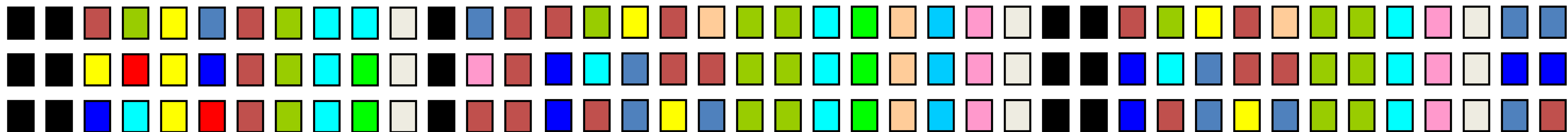
time

Continuous Aggregation

How many red colored boxes are there?

8

1 minute wide window



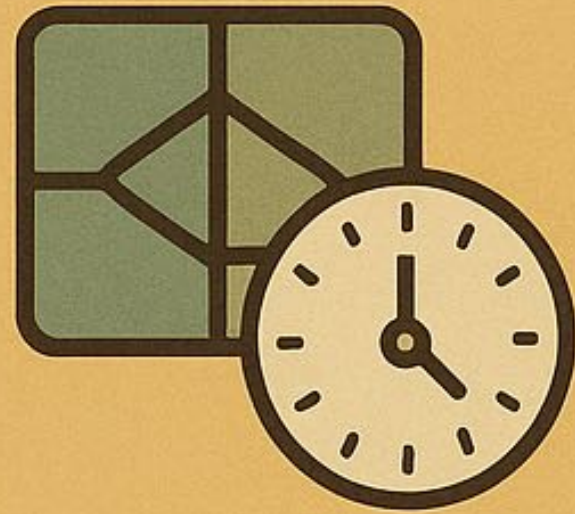
time

SYNTAX EXTENSION



- bolt-on clauses (TUMBLE/HOP/SESSION)
- time attributes & gaps

SEMANTICS EXTENSION



- membership rules
- overlap & triggers
- late data & corrections

PRAGMATICS MISMATCH



- user need vs system behavior
- watermarks/latency knobs
- same intent $\neq$ same answer

Windows extend the language and the meaning – but misalign with how people use results.

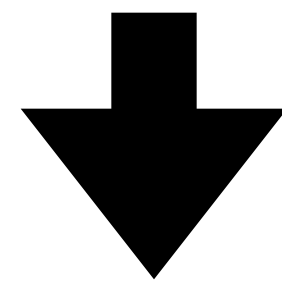
Windows

What is wrong with them?

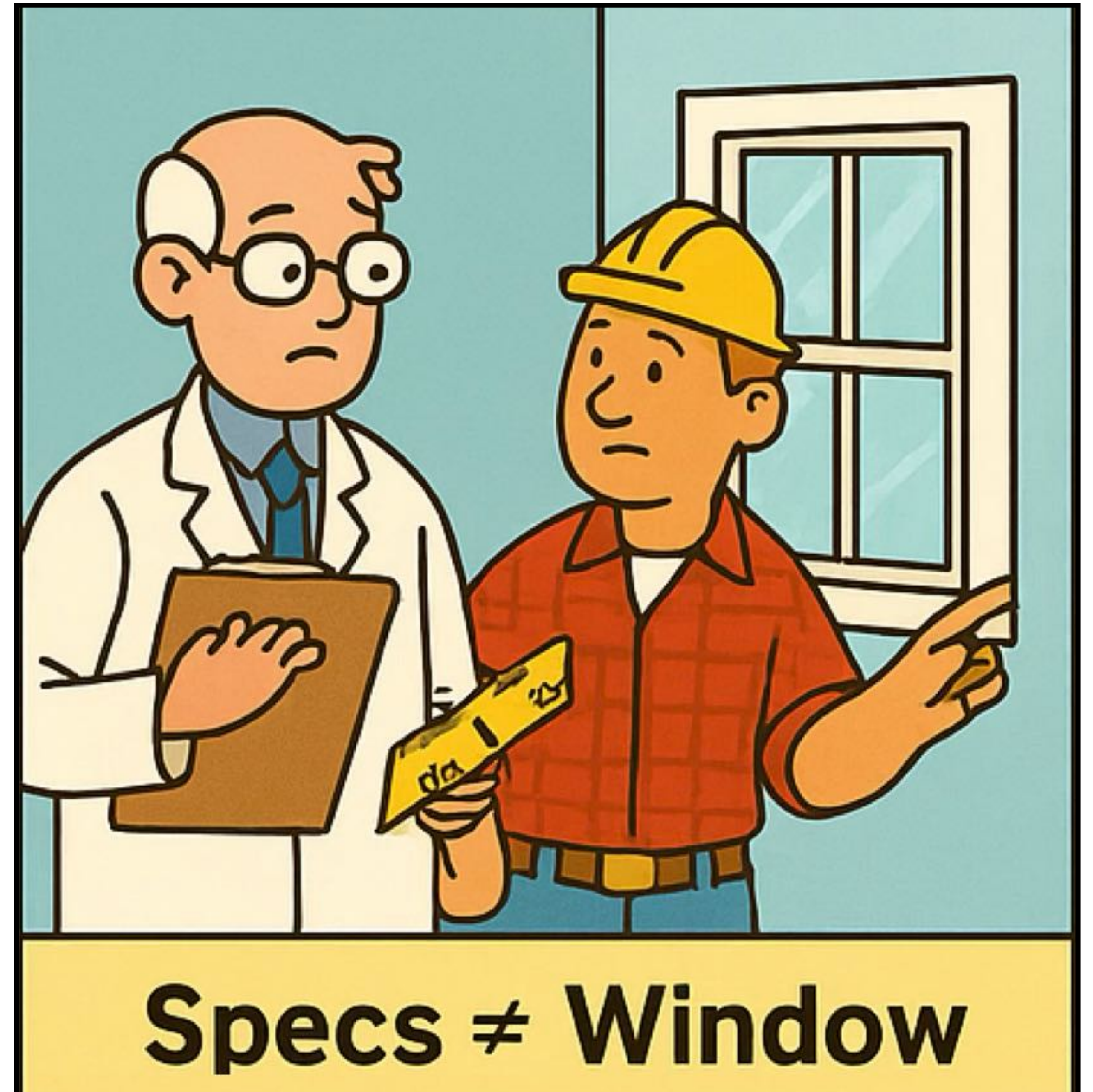
Smart-Factory Scenario

Problem specs rarely define a window

We want to detect thermal stress:
whenever the average temperature
per machine over a **5-minute
tumbling window** exceeds 80 °C,
emit a record.



```
SELECT did, AVG(temp_c) AS avg_temp_c  
FROM iot_readings  
GROUP BY TUMBLE(ts, INTERVAL '5'), did  
HAVING AVG(temp_c) > 80;
```

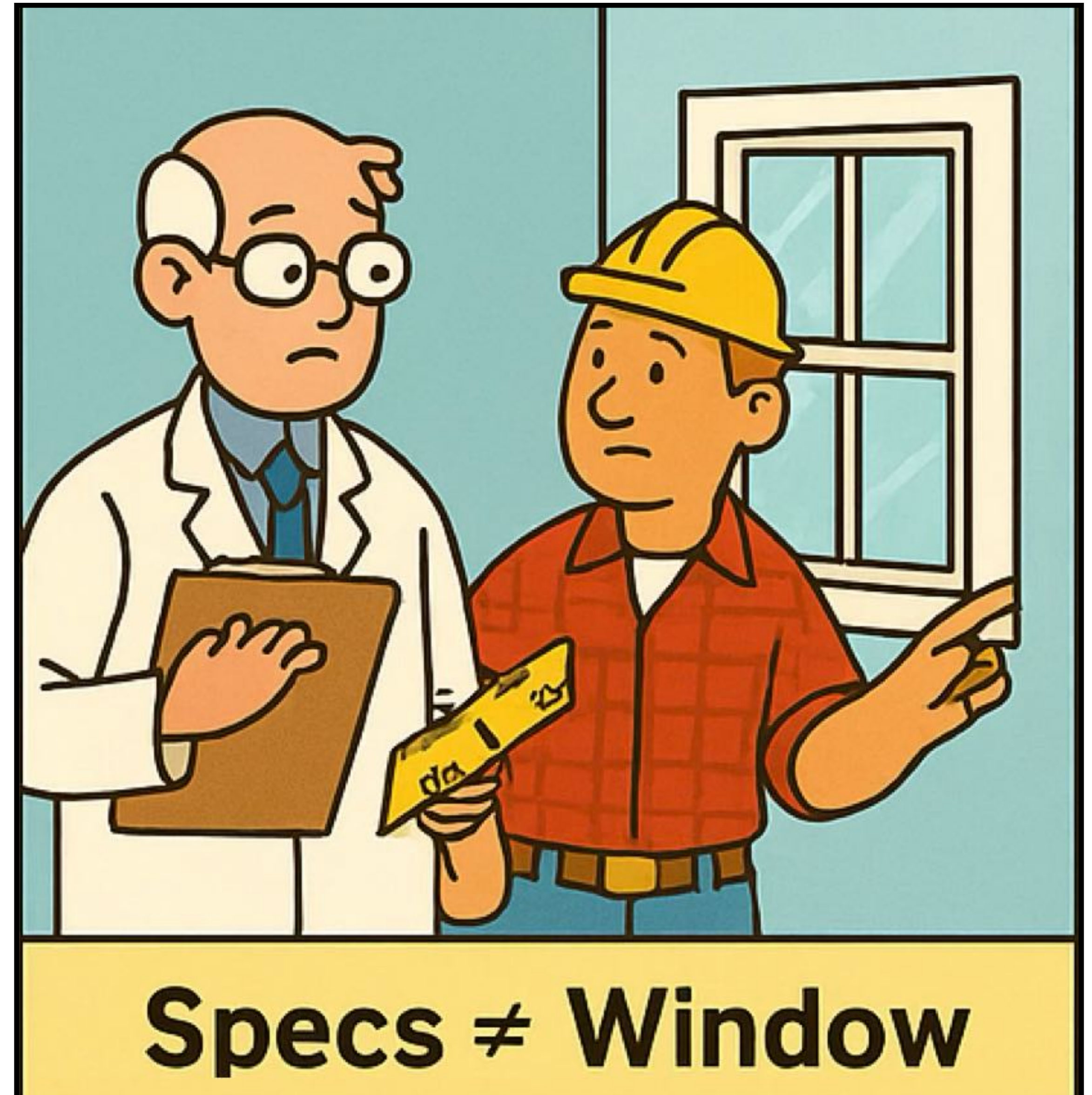


Smart-Factory Scenario

Problem specs rarely define a window

A **thermal-stress alert** when the mean **temperature** is “**high**” while the machine runs **steadily**, and we must notify quickly enough to act.

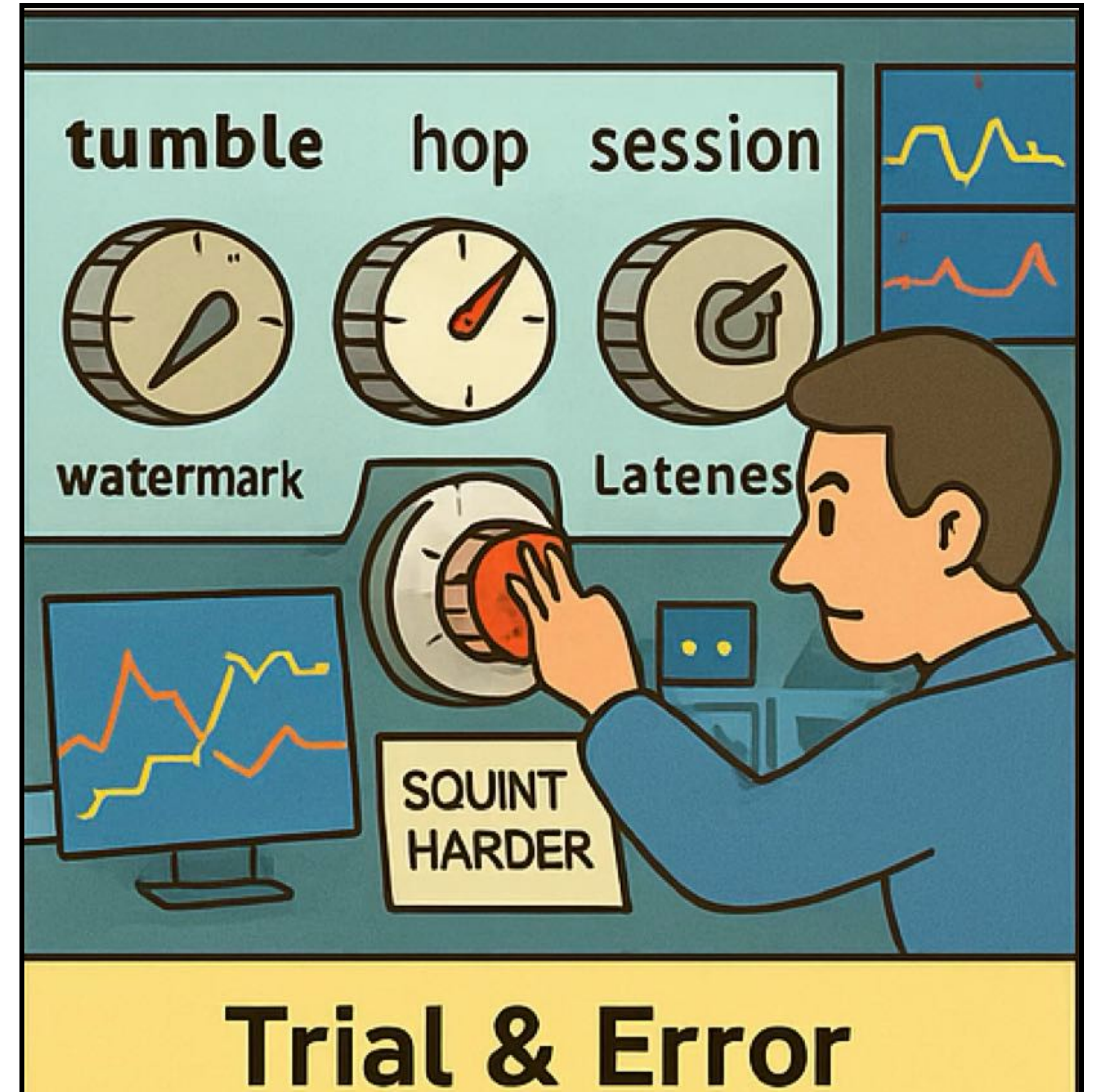
- **Stability** → Only evaluate while regime is steady (bounded rpm change, low temp variance).
- **Sufficient evidence** → Require enough samples to get a CI on mean temp (e.g., $N \approx 150$ at 0.5 Hz).



Smart-Factory Scenario

Users learn differences empirically

- First users they solve their information need (it may be too noisy, too much data)
- They get it smoother but overlapping (duplicates surprise users)
- They “Learned” tweaks (evidence + stability)



Example of Trial and Error

Using Flink SQL

```
-- 1) Naive (too noisy): 1-minute tumbling window
SELECT
  device_id,
  AVG(temp_c) AS avg_temp
FROM iot_readings
GROUP BY TUMBLE(ts, INTERVAL '1' MINUTE), device_id
HAVING AVG(temp_c) > 80;
```


Example of Trial and Error

Using Flink SQL

```
-- 1) Naive (too noisy): 1-minute tumbling window
```

```
SELECT
```

```
device_id,
```

```
AVG(temp_c) AS avg_temp
```

```
FROM iot_readings
```

```
GROUP BY device_id
```

```
HAVING avg_temp > 80;
```

```
-- 2) Overlapping windows (duplicates): 5-min window,  
1-min hop
```

```
SELECT
```

```
device_id,
```

```
AVG(temp_c) AS avg_temp
```

```
FROM iot_readings
```

```
GROUP BY HOP(ts, INTERVAL '1' MINUTE, INTERVAL '5'
```

```
MINUTE), device_id
```

```
HAVING avg_temp > 80;
```

Example of Trial and Error

Using Flink SQL

```
-- 3) "Learned" tweaks: evidence + stability (5-min  
1- tumbling)  
SE  
SELECT  
    device_id,  
    AVG(temp_c) AS avg_temp,  
    COUNT(*) AS samples  
FR  
GR  
MI  
HA  
FROM iot_readings  
GROUP BY TUMBLE(ts, INTERVAL '5' MINUTE), device_id  
HAVING AVG(temp_c) > 80  
    AND COUNT(*) >= 150 -- enough samples  
(evidence)  
    AND (MAX(temp_c) - MIN(temp_c)) <= 15; -- crude  
stability check
```

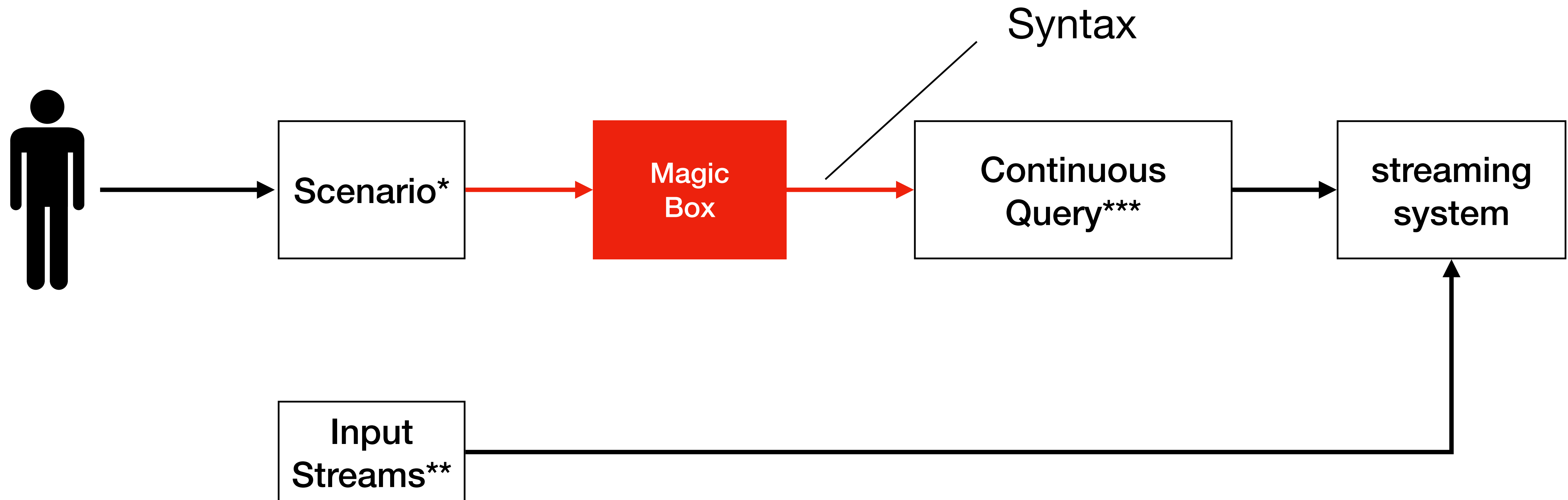

**How can we make
them better?**
maybe using AI?





Hide Windows From Their Users

A Magic Box



* Natural Language

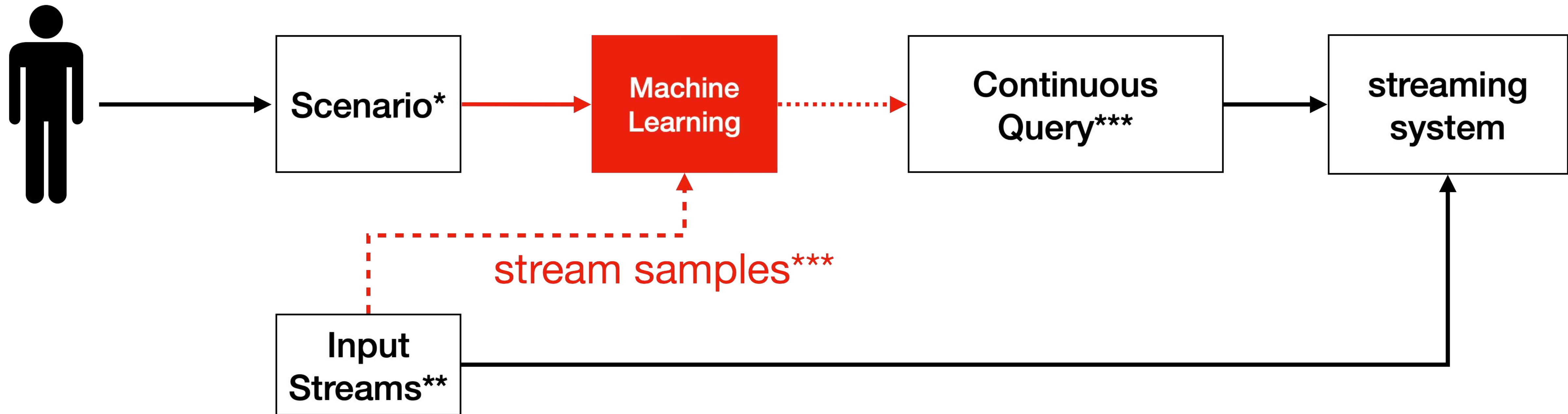
** Metadata Available

***Window Based



Hide Windows From Their Users

Is ML the Magic Box?



* Natural Language

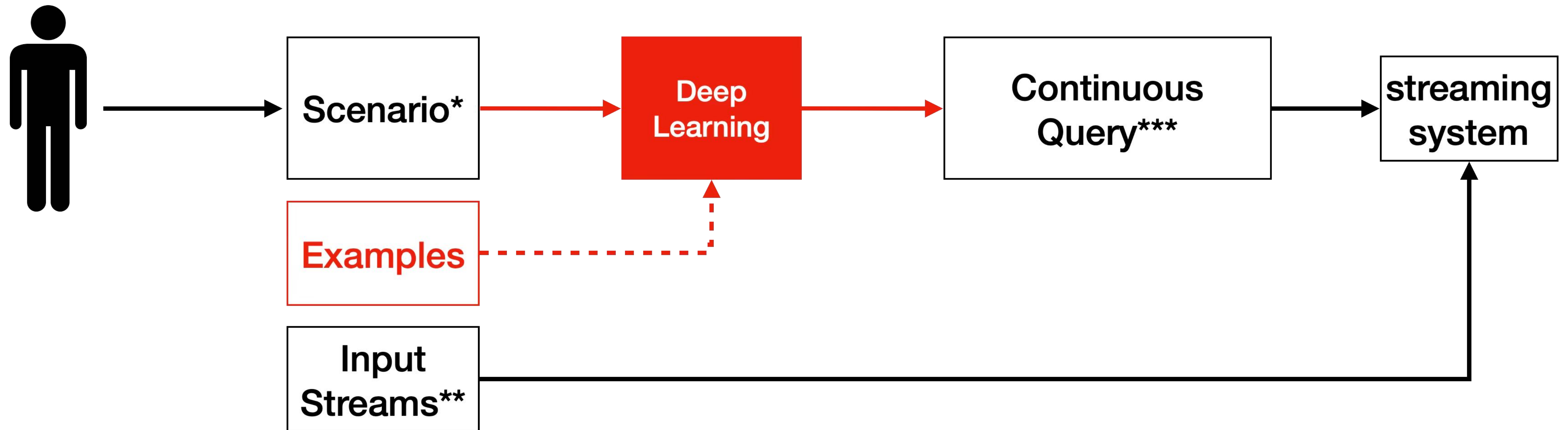
** Metadata Available

***Window Based



Hide Windows From Their Users

Is DL the Magic Box?



* Natural Language

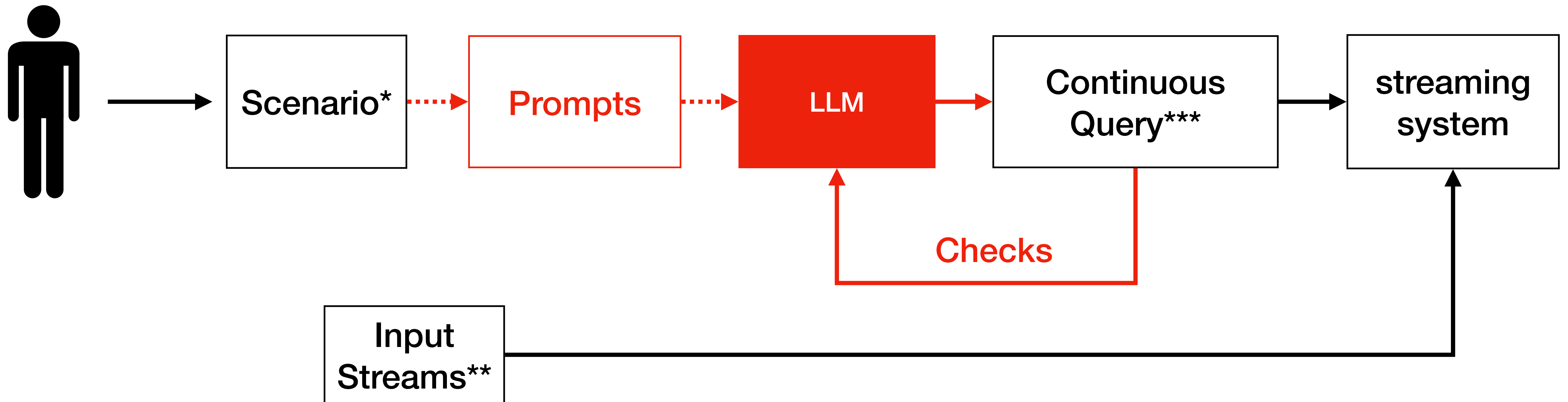
** Metadata Available

***Window Based



Hide Windows From Their Users

Are LLMs the Magic Box?



* Natural Language

** Metadata Available

***Window Based

Summary

Human expert

- Problem specs rarely define a window
- Users learn differences empirically
- Long-lived streams break assumptions
- Implicit correctness leads to silent failures

ML-Based

- As a classification: it needs labels
- As clustering, it needs requires stream samples;
- it may struggle on novel cases.

DL-Based

- task-specific;
- Data-hungry;
- opaque;
- hard to constrain;
- fails under concept drift.

LLM Based

- No guarantees;
- may invent operators;
- brittle under drift;
- but possible improvements via prompting techniques



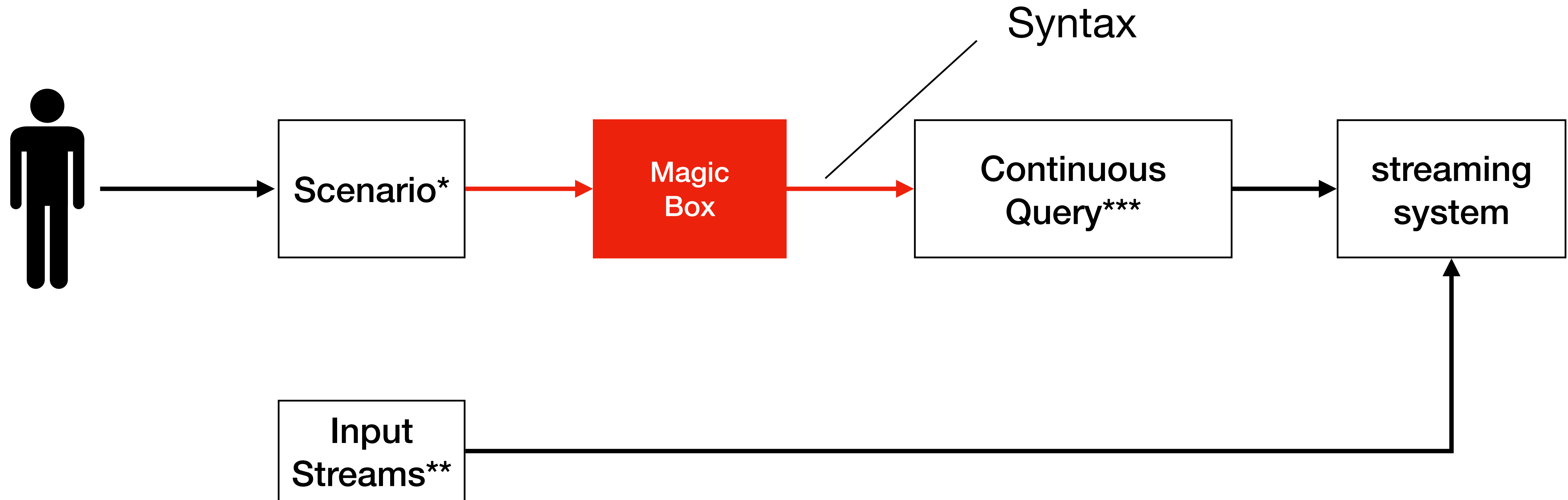
WE NEED TO GO

DEEPER



Hide Windows From Their Users

A Magic Box



* Natural Language

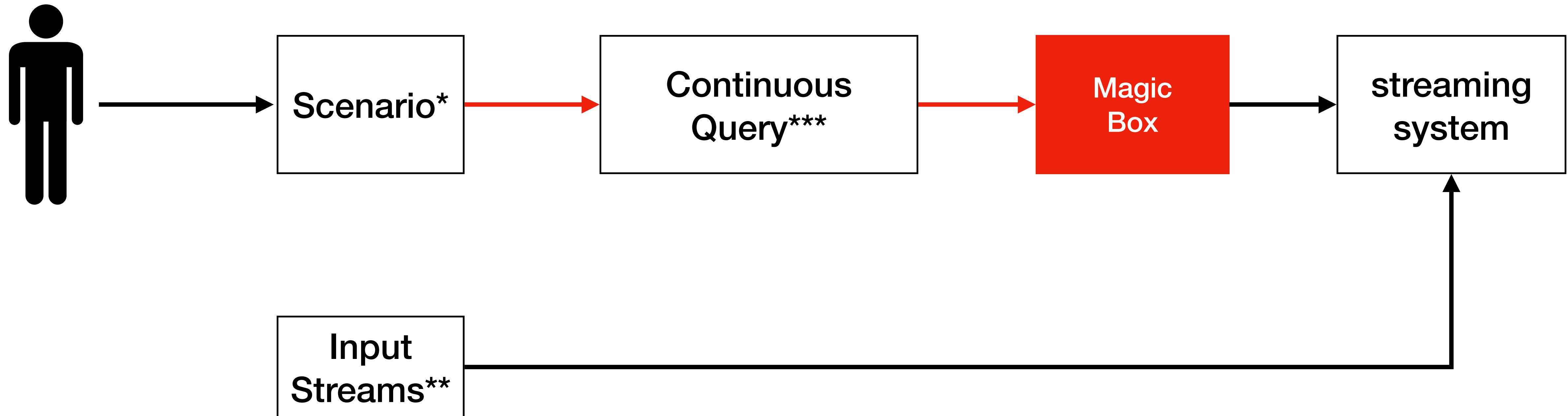
** Metadata Available

***Window Based



Hide Windows From Their Users

A Magic Box



* Natural Language

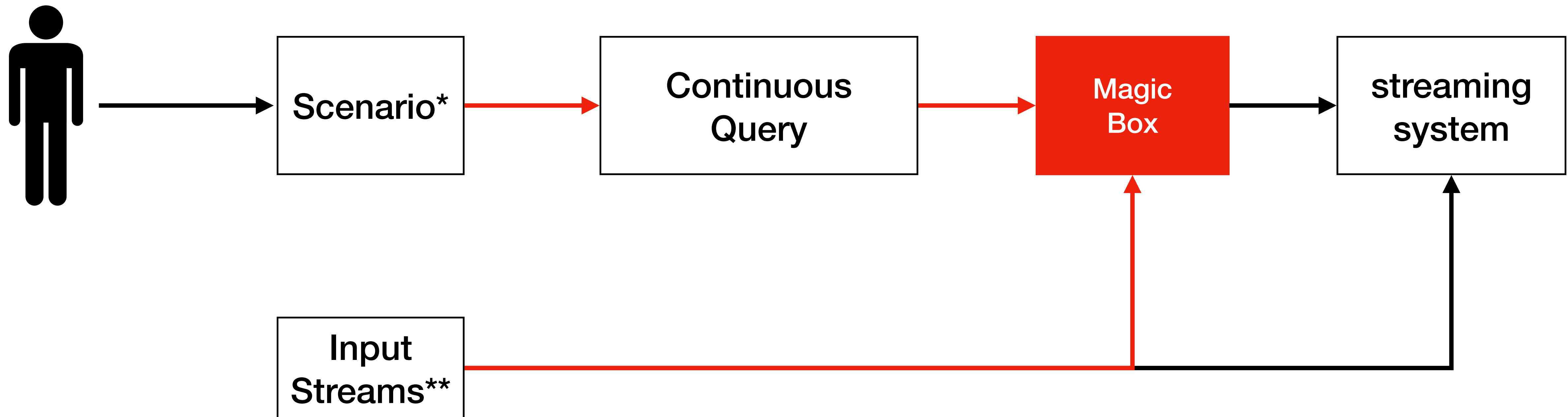
** Metadata Available

***Window Based



Hide Windows From Their Users

A Magic Box

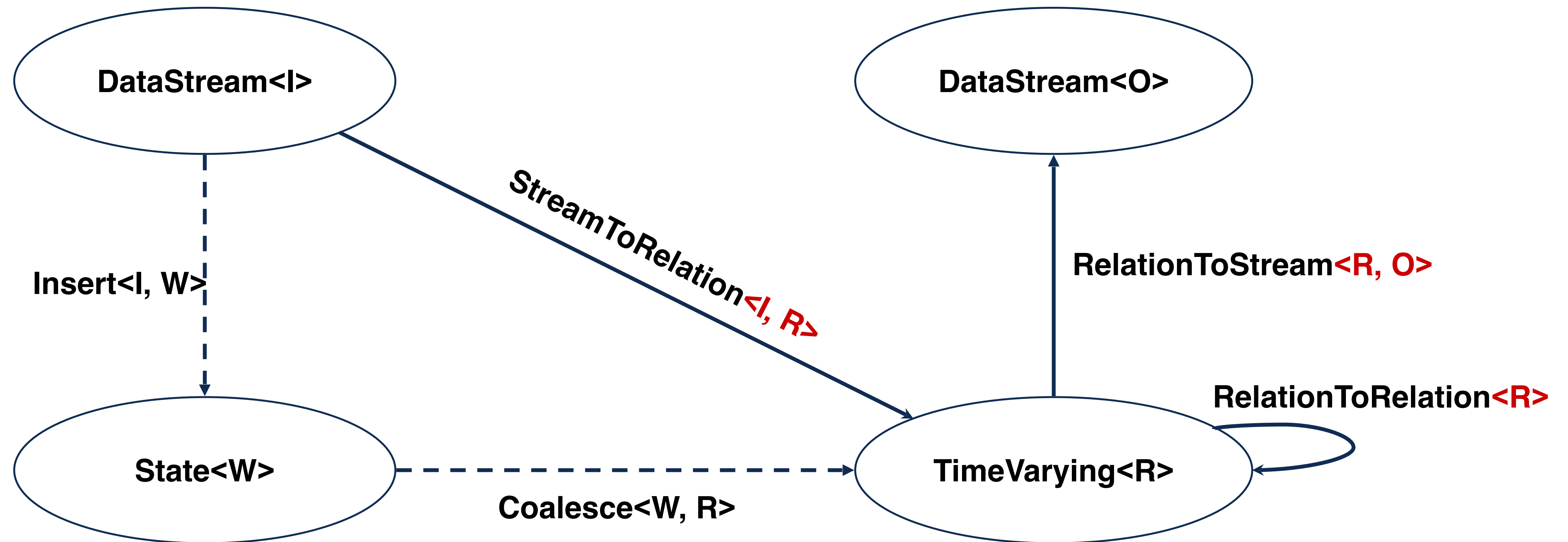


* Natural Language

** Metadata Available

Generalising CQL

PolyFlow Repo at : github.com/riccardotommasini/polyflow-examples/

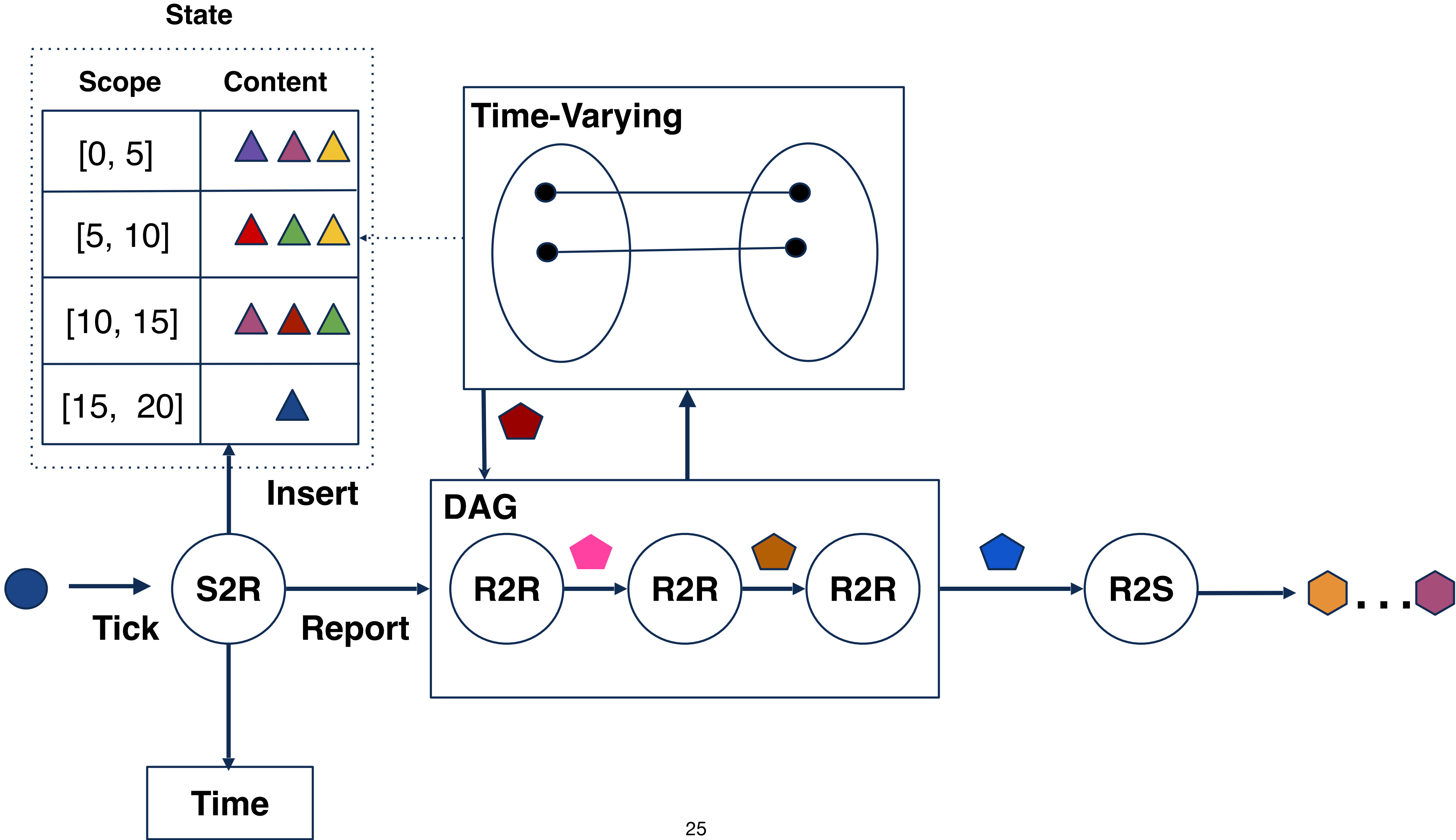


PolyFlow

In practice

Legend

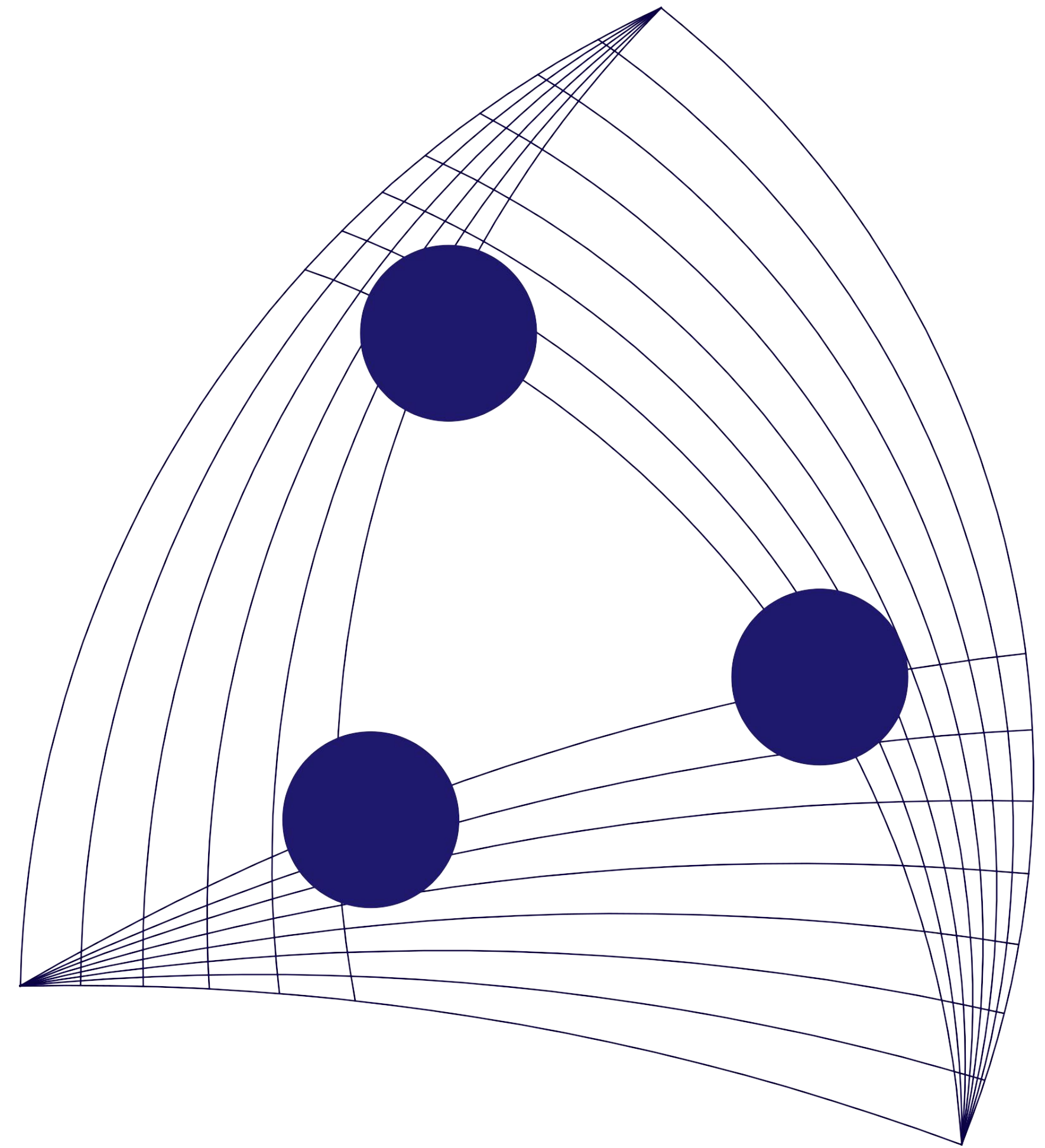
- = I
- ▲ = W
- ⬠ = R
- ⬡ = O



Working on PolyFlow

Old Logo -->

- we have a number of tasks to be done with the library (java coding required)
- creating new features
 - real-time UI using Vaadin
 - Off Heap Memory Management
 - Streaming Query Optimisation
 - Streaming SQL with LLMs

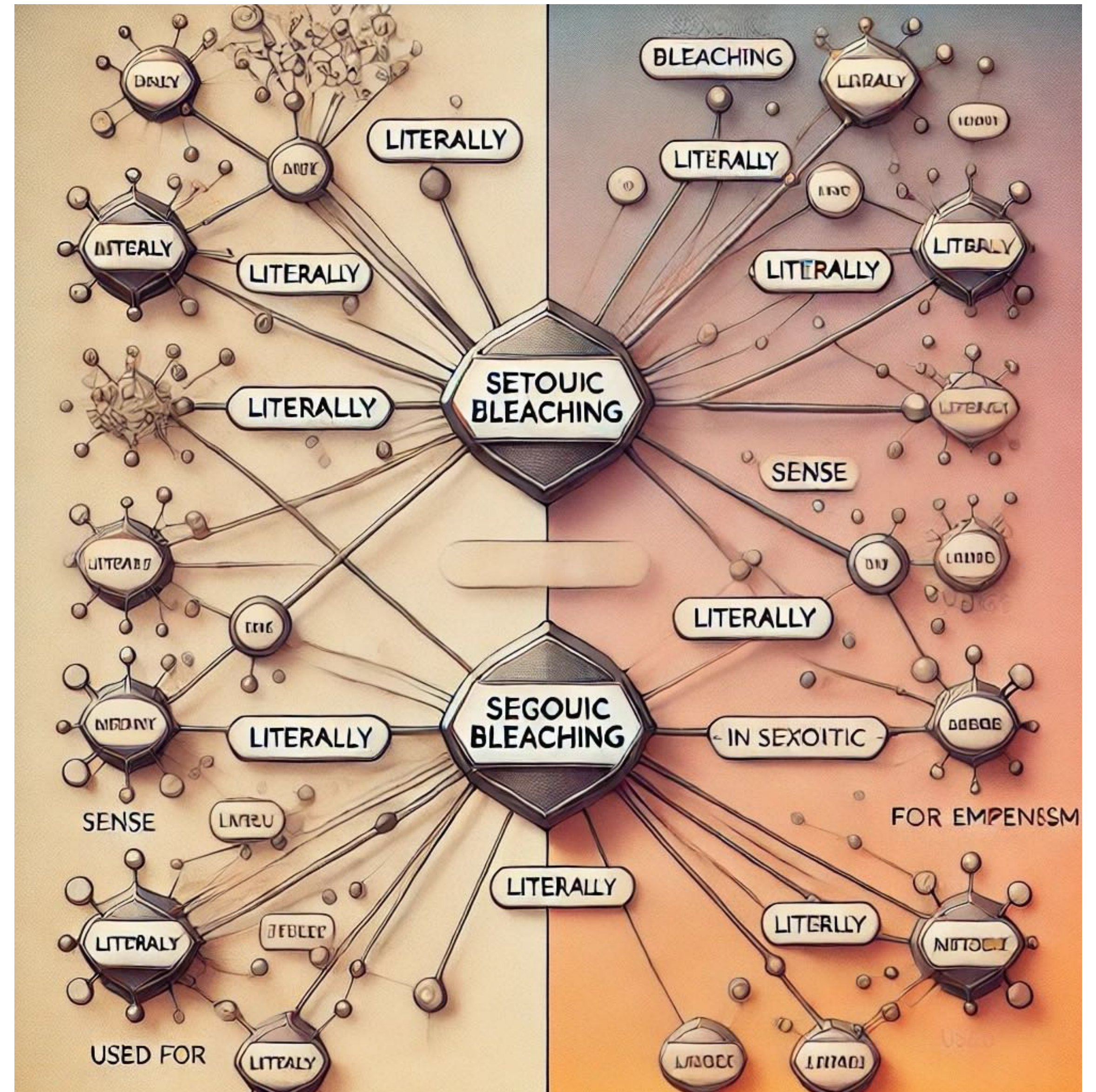


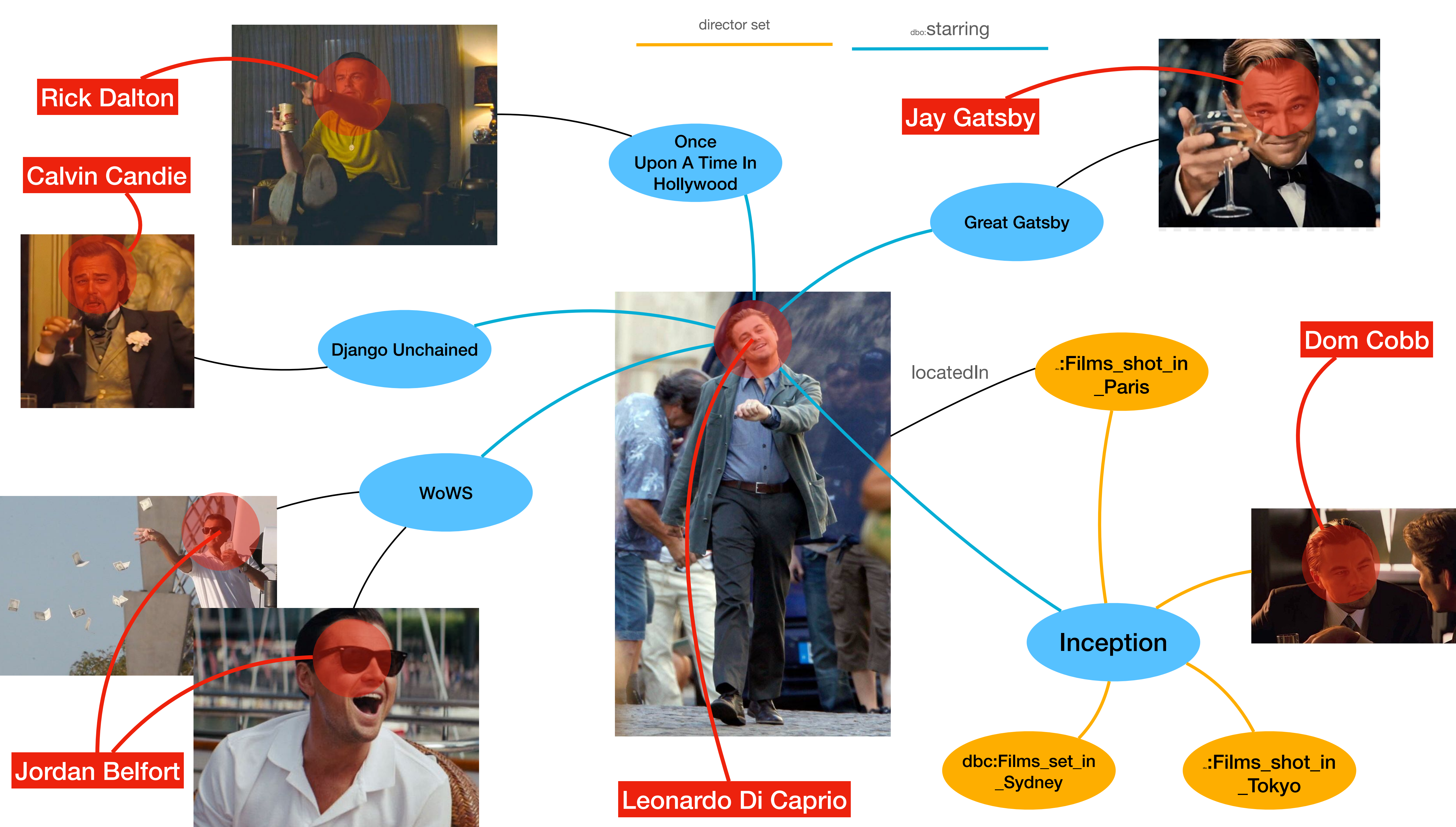
RSPJ

About PolyFlow

- PolyFlow Repo at : github.com/riccardotommasini/polyflow
- Quick Start at : <https://github.com/riccardotommasini/polyflow-quickstart/>
- Examples Repo at : github.com/riccardotommasini/polyflow-examples/
- Recent Example of work: <https://github.com/semlanghi/playtwin/>

Knowledge and Graphs





Dissection of an Internet Meme

a) Origin

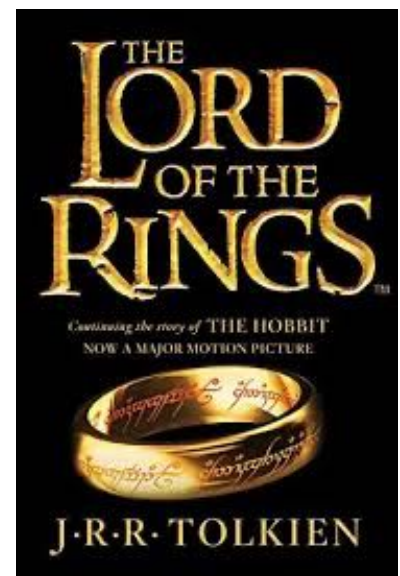


The picture is from **Peter Jackson**'s 2001 movie "LOTR: The fellowship of the ring" where **Sean Bean** played "**Boromir**"



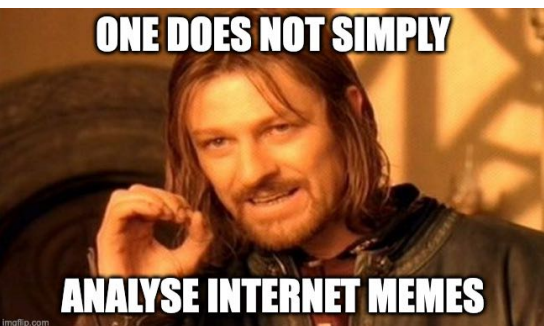
b) Lore

The **book series** "Lord Of The Ring" by **J. R.R. Tolkien** features the character Boromir and the fictional region of **Mordor**



d) Meme

One does not simply *analyse internet memes*



c) Template

One does not simply walk into Mordor



e) Frame

Difficulties_in_action: factor(s) negatively affecting the ability to achieve objective



MedigKG

Knowledge Infused Deep Learning for Image Analysis

